

Final Review

Name: _____ ID: _____ Group: 212/ 1

This guide will help you prepare for your final exam. You don't have to submit anything, and it won't count as an activity.

In addition to this guide, you'll need to complement your study for the final exam with your **quizzes from each partial, your partial exams, the lab practices we did and the class notes**. Additionally, you may consult the next book, "Chemistry: An Introduction to General, Organic and Biological Chemistry", by Karen C. Timberlake, eleventh edition.

List of topics

Carbon (Its importance in organic chemistry, hybridization)

Differences between organic and inorganic materials

Alkanes (Uses, sources, properties, nomenclature and stoichiometry problems)

Reactions: Complete and incomplete combustion, halogenation and cracking. Processes: Fractional Distillation

Alkenes (Uses, sources, properties, nomenclature and isomers)

Reactions: Hydrogenation, Hydration, Halogenation, Hydrohalogenation, Polymerization (Only theory)

Alkynes (Uses, sources, nomenclature)

Reactions: Hydrogenation

Aromatics (Main compounds, properties, uses, sources, nomenclature (o,m,p indicators too))

Reactions: Halogenation (Chlorination and Bromination), Nitration, Sulfonation, Friedel-Crafts (Alkylation and acylation)

Alcohols (Uses, sources, properties, nomenclature)

Reactions: Fermentation, Oxidation and Dehydration

Ethers (Uses, nomenclature)

No reactions

Aldehydes (Uses, sources, properties, nomenclature)

Reactions: Oxidation and Reduction

Ketones (Biochemistry, uses, sources, properties, nomenclature)

Reactions: Reduction

Carboxylic acids (Uses, sources, properties, nomenclature)

Reactions: Neutralization, Esterification, Reduction

Esters (Uses, sources, properties, nomenclature)

Reactions: Hydrolysis (Saponification)

Amines (Uses, sources, properties, nomenclature)

No reactions

Amides (Uses, sources, properties, basic nomenclature)

No reactions

Biomolecules (Only lipids and carbohydrates)

Part I

Fill in the blanks

Organic chemistry is the study of compounds that contain primarily _____ and _____ atoms, but aside from those they usually contain too _____, _____, _____, _____ and even halogens like _____, _____ and _____.

Carbon is such an important element because as it is in the _____ family it has 4 _____ electrons, allowing it to bond _____ times and create a vast array of molecules with different chemical properties. Another characteristic is that C-C bonds are _____ enough to be stable, but at the same time not too strong making it possible for them to break and rearrange in new patterns.

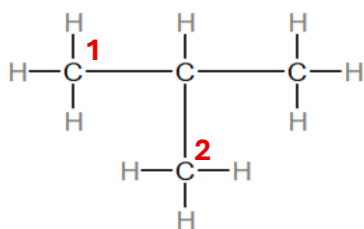
According to the electron configuration of carbon, its 4 valence electrons can be found in orbitals _____ and _____. Those orbitals usually combine to create stable bonds. When we combine the s orbital with all three p orbitals it results in a _____ hybridization, creating _____ hybrid orbitals.

Each single bond is known as a _____ bond and it is the strongest type of bond. In every double bond we have one _____ bond and one _____ bond. Finally, in each triple bond we have one _____ bond and two _____ bonds.

Complete the next table

Hybridization	Type of bond(s)	Molecular geometry	Bond angle	Create one example
sp^1	Sigma and pi			
			120°	
		Tetrahedral		

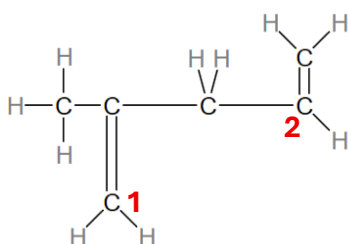
For each example, write the name of the molecule and indicate the type of hybridization for the carbons.



Hybridization (C1):
Hybridization (C2):

Name

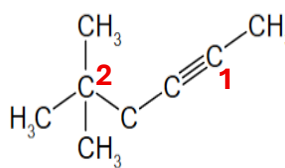
Sigma bonds:
Pi bonds:



Hybridization (C1):
Hybridization (C2):

Name

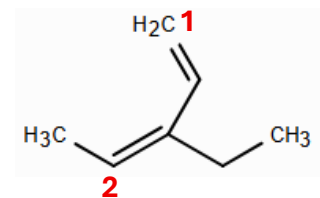
Sigma bonds:
Pi bonds:



Hybridization (C1):
Hybridization (C2):

Name

Sigma bonds:
Pi bonds:



Hybridization (C1):
Hybridization (C2):

Name

Sigma bonds:
Pi bonds:

Part II

Complete the next table with the difference between organic and inorganic compounds

Property	Organic	Inorganic
Elements present		Most metals and non-metals
Particles	Mostly molecules	
Melting point	Low	
Boiling point		High
Conductivity	Bad conductors	
Flammability		
Solubility in water	Not soluble, unless a polar group is present	

Give examples of at least 4 common organic compounds and their molecular formulas

Decide whether the next substances are organic or inorganic

Sodium hydroxide

Chloroform

Margarine

Paraffin

Urea

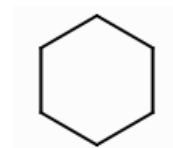
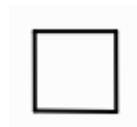
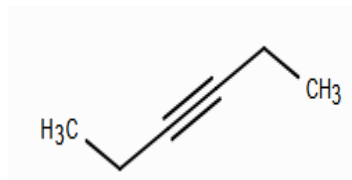
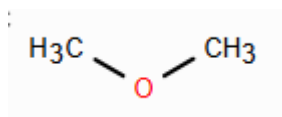
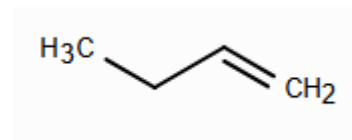
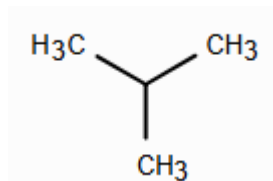
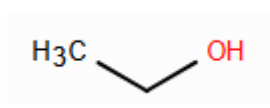
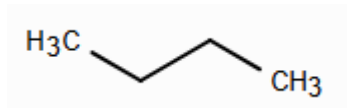
Table salt

Sulfuric acid

Ethanol

What is the definition of an isomer?

Indicate which of the following molecules are isomers



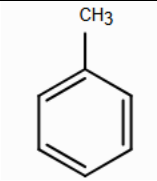
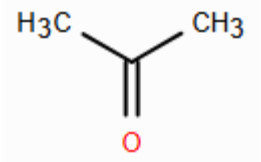
Propose the structure of an isomer of 2-methyl-1,3-butadiene and name it

What is the difference between constitutional isomers and stereoisomers?

Part III

Complete the next table with the information about each functional group we saw during the semester.

Functional group	General formula	Example (Structure and name)	Properties	Uses
Alkanes	C_nH_{2n+2}			
Alkenes				As feedstock for the chemical industry

	C_nH_{2n-2}		Low polarity and insoluble in water	
		 Toluene		
Alcohols				
Ether			Quite inert but react violently in air. They are mostly insoluble in water	
	$R-CHO$			
		 Propanone		
Carboxylic acids				
	$R-COO-R'$			

Amines				In drugs and vitamins. As starting material in various synthetic processes. In synthetic dyes.
	R-CONH ₂			

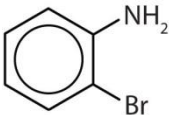
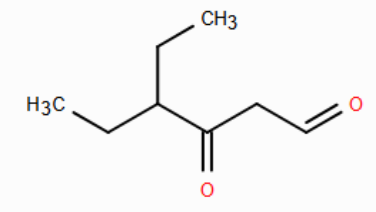
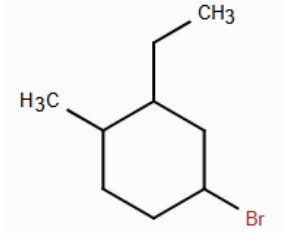
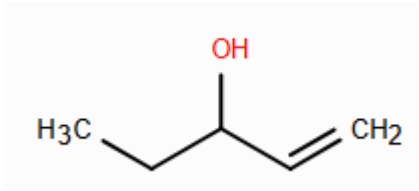
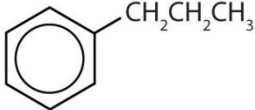
Part IV

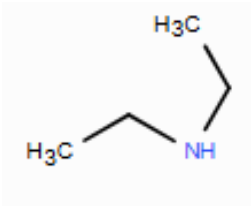
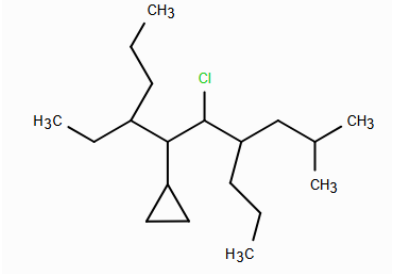
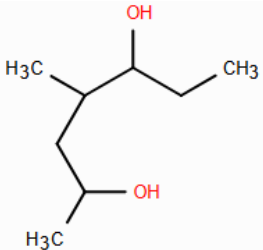
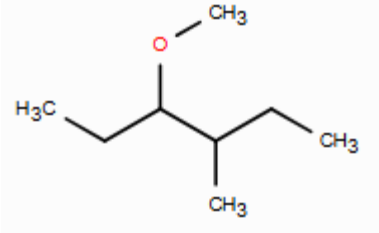
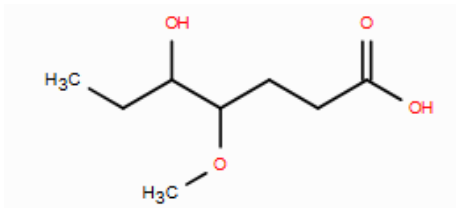
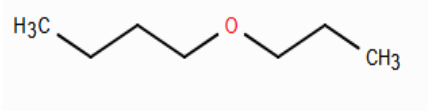
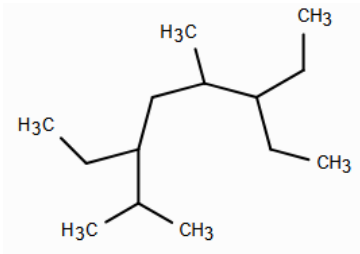
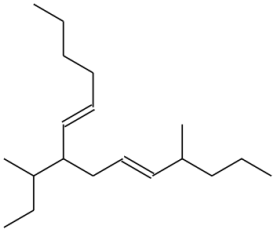
Nomenclature

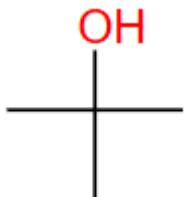
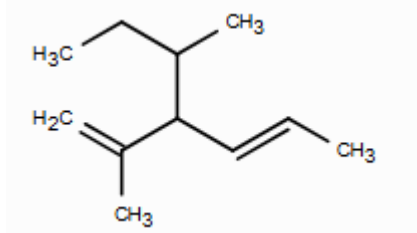
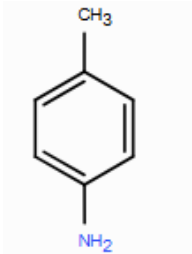
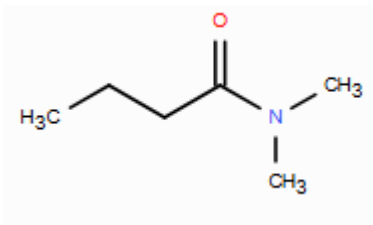
The priority we will use is:

Carboxylic acid > Ester > Amide > Aldehyde > Ketone > Alcohol > Amine > Alkenes/Alkynes > Haloalkanes/Ethers

For each compound write the name following IUPAC rules or draw the Kekule or skeletal structure.

		
	3,3-dimethyl-5-phenyloctane	
		
1-penten-2,3,4-triol		Trichloromethane
		
	4-isopropyl-3-(2-methylpropyl)-5-oxoheptanal	

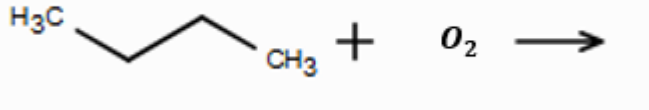
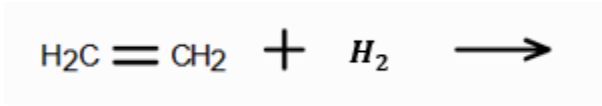
		
1,2-dimethylcyclopropane		
		
		N,N-dimethylethanamine
		
3-penten-1-yne		5-phenyl-2-octene
		
3-methylbutyl pentanoate		Nitrobenzene
		
	Ethyl propanoate	

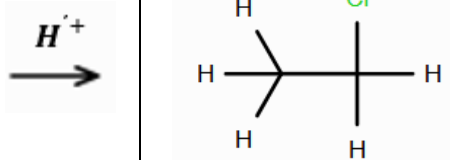
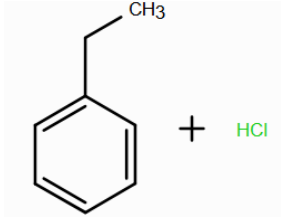
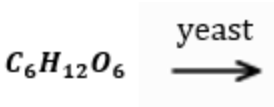
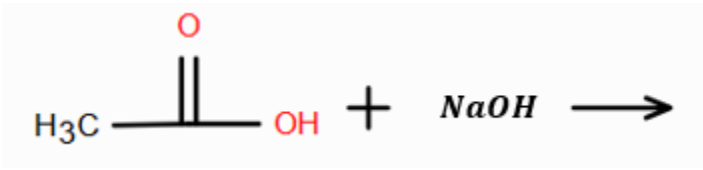
		
		3,4-heptanediol
		
p-secbutylbenzoic acid		

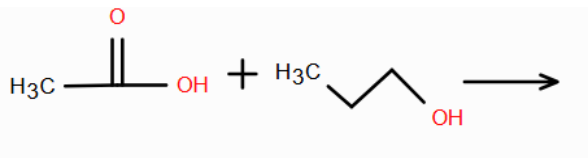
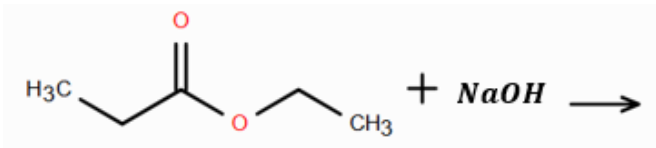
Part V

Reactions

For each exercise write the type of reaction and complete the names and structures of reagents and products. Also, indicate if there is need for a catalyst.

1. Complete combustion 	
2. Incomplete combustion (With carbon monoxide as product)  2-methylbutane molecular oxygen	
3.	2-chloro propane hydrogen chloride
4. 	

5. Hydration of propene	
6. Hydrohalogenation	 The diagram shows the hydrohalogenation of isobutylene with H^+ to form 2-chloro-2-methylbutane. The reactant is isobutylene, and the product is 2-chloro-2-methylbutane, where the chlorine atom is highlighted in green.
7.	 The diagram shows toluene (a benzene ring with a CH_3 group) reacting with HCl (highlighted in green).
8.	 The diagram shows the chemical formula $C_6H_{12}O_6$ reacting with yeast to produce an unknown product.
9. Complete oxidation of 1-propanol	
10. Complete oxidation of 2-butanol	
11. Complete reduction of 2-methylbutanoic acid	
12.	 The diagram shows the chemical reaction of acetic acid (CH_3COOH) with sodium hydroxide ($NaOH$) to form an unknown product. The carboxylic acid group is highlighted in red.

13. 	
14. 	

Part VI

Biomolecules

What are the four main biomolecules?

Lipids are ideal components of cell structure due to their _____. The human body also uses them as sources of energy, that's the foundation of a _____ diet.

Fatty _____ are one of the primary type of lipids, they are long _____ and can be saturated or _____. If we combine three fatty acids with one molecule of _____ we will get a _____. In a fatty acid, the hydrocarbon tail must be between ____ to ____ carbons.

What are some functions of fatty acids and where can we obtain them from?

The other three types of lipids are _____, _____ and _____.

_____ are responsible for short term storage of energy in living organisms and they make up the main structural components in _____. As the name suggests carbohydrates are made of _____, _____ and oxygen atoms.

Carbohydrates can be divided into two categories: _____ and _____, which are also known as complex carbohydrates. Cellulose, glycogen and _____ are part of that last group.